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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS257 - HIGH PERFORMANCE COMPUTING

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1 A multiprocessor machine which is capable of executing multiple instructions on multiple data sets.**
a) SISD b) SIMD c) MIMD d) MISD
2. **CO1 Utilization rate of resources in an execution model is known to be its?**
a) adaptation b) efficiency c) dependability d) flexibility
3. **CO2 Suppose we run a parallel program with a fixed number of processes/threads and a fixed input size, and we obtain an efficiency E. Suppose we now increase the number of processes/threads that are used by the program and find a corresponding rate of increase in the problem size so that the program always has efficiency E. Which of the following suits such a program?**
a) Not scalable b) Scalable c) May be scalable d) Optimizable
4. **CO1 FADD, FMAD, FMIN, FMAX are _____ supported by Scalar Processors of NVIDIA GPU.**
a) 32-bit ieee floating point instructions b) 32-bit integer instructions
c) 16 Bit Binary instructions d) 16-bit floating point instructions
5. **CO1 CUDA supports _____ in which code in a single thread is executed by all other threads.**
a) tread division b) tread termination c) thread abstraction d) none of the above

PART - B

(5x 3 = 15 Marks)

6. **CO1** Define the taxonomy of parallel computing paradigms.
7. **CO2** Give an example of a vector operation.
8. **CO3** Compare point-to-point and collective communication in MPI.
9. **CO4** Indicate how sparse matrix computation benefits from data compression.
10. **CO4** Predict the output behavior of a CUDA program using dynamic parallelism.

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) Explain the architecture of a multicore processor with a neat diagram. (5)
Discuss the advantages and challenges in designing multicore systems.
- CO1** ii) Compare shared memory and distributed memory systems with examples, advantages, disadvantages, and use cases. (5)
12. **CO2** A program runs in 10sec in computer A which has 2GHz clock. A computer designer builds a computer B that runs the same program in 6 sec. But there is a need of substantial increase of clock rate that causes computer B requiring 1.2 times as many clock cycles as computer A. Evaluate the clock rate should we tell the designer to achieve target? (10)
13. **CO3** i) Explain the concept of wave front parallelization in OpenMP with the help of a matrix computation example. (5)
- CO3** ii) Implement the Jacobi iterative solver using OpenMP, and annotate the key sections of code where parallelism is introduced. (5)
14. **CO3** i) Explain MPI point-to-point communication with example code for MPI_Send and MPI_Recv. (5)
- CO3** ii) Assess the efficiency of a parallel Jacobi algorithm implemented in both OpenMP and MPI, based on execution time, scalability, and memory usage. (5)

15. **CO4** i) Explain the principles of scalable parallel execution in modern multicore and many core systems. (5)
- CO4** ii) Describe how MU programming supports concurrent execution with an example architecture. (5)
16. **CO4** Design a machine learning workflow that uses parallel patterns to accelerate training and inference on large datasets. (10)
17. **CO3** i) Implement a matrix multiplication algorithm that scales across multicore systems. (5)
- CO3** ii) Create a strategy using decomposition and pattern recognition to parallelize a simulation workflow. (5)
